EDUVISIONAI : AI-POWERED HISTORY E-LEARNING SYSTEM FOR VISUALLY IMPAIRED O-LEVEL STUDENTS IN SRI LANKA

25-26J-438

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specializing in

Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

..... 2025/8/28

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ABSTRACT

This research project proposes the development of a voice-based Question-Answering (QA) system designed to support visually impaired O-Level students in learning English-language History in Sri Lanka. The system leverages speech recognition, text-to-speech synthesis, and AI-powered answer evaluation to provide an interactive and accessible learning environment. Students engage in quizzes generated by the system, receiving real-time feedback and adaptive learning support.

The project addresses a critical gap in inclusive education: while several AI-based tutoring systems exist, few focus on curriculum-specific content for visually impaired learners, particularly in the humanities. The QA system will be trained on the official Sri Lankan O-Level History syllabus, ensuring accuracy, relevance, and alignment with educational standards.

Adopting an Agile development methodology, the project emphasizes iterative refinement, incorporating feedback from visually impaired students and history educators. Core functionalities include curriculum-aligned QA, adaptive difficulty adjustment, performance tracking, and accessibility-optimized voice interaction. A detailed feasibility study confirms the project's technical, operational, and economic viability, while multi-level testing ensures usability, reliability, and real-time responsiveness.

By combining AI-driven assessment with voice-based interaction, this system aims to enhance inclusive learning, foster student engagement, and enable visually impaired learners to independently access, understand, and practice curriculum content. The proposed platform represents a significant step toward accessible, technology-enabled education in Sri Lanka, offering a model that can be adapted to other subjects and educational contexts in the future.

KEYWORDS

Voice-Based Learning

AI-Powered Question-Answering

Visually Impaired Students

Inclusive Education

O-Level History Curriculum

Speech Recognition

Text-to-Speech (TTS)

Adaptive Learning

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
QA	Question and answering
VI	Visually Impaired
TTS	Application Programming Interface
UI	User Interface
O/L	Ordinary Level
NLP	Natural Language Processing
WCAG	Web Content Accessibility Guidelines
UAT	User Acceptance Testing
API	Application Programming Interface
SDK	Software Development Kit

Table 1: List of Abbreviations

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1. INTRODUCTION

1.1 Background and Literature

Visually Impaired (VI) students in Sri Lanka are still facing long-lasting educational struggles despite the free education system in Sri Lanka. Specialist materials they need such as Braille books, tactile diagrams, and sound materials are still scarce, while equipment and teacher training in VI pedagogy are often inadequate [1]. Even when schools have special or comprehensive programs, they often face not enough resources and some lingering attitudes. For example, research points to the fact that VI students tend to "rely on others" to use fundamental tools; an ordinary calculator lacking voice capability becomes inaccessible, and so "talking" calculators are created [2]. The challenges emphasize how important it is to bring easy-to-use, tech-driven help into everyday classrooms.

Voice-controlled interfaces offer a solution to these obstacles in the sense that they offer sound access to learning material and examinations. Speech technologies like text-to-speech, speech-to-text, and voice commands help visually impaired students access learning materials more easily. In Sri Lanka, early studies have explored this potential. One such study proposed a computer-based, highly automated, voice-controlled online examination system where students give answers orally to questions and receive immediate feedback [3].

Globally, intelligent voice-based e-learning platforms have been shown to be of great benefit to visually impaired students, particularly in distance learning environments [4]. Such platforms use AI-powered speech synthesis and recognition to create interactive, fair, and adaptive learning experiences. Delivering content and assessments via voice not only improves accessibility but also promotes learner independence and engagement, helping to address both academic and social integration challenges faced by VI students

1.2 Research Gap

There have been several intelligent tutoring tools created to support blind students, though most focus primarily on math or science content. For example, Maćkowski *et al.* (2018) built an accessible interactive math tutoring platform: that simplifies complex problems into smaller steps, verifies each solution, and provides audio descriptions of formulas along with detailed feedback [5]. In a related system, Maćkowski *et al.* (2022) use a tablet with audio-tactile graphics to teach math to blind learners. This platform automatically selects relevant exercises and offers spoken explanations of charts and formulas, which has been reported to enhance students' understanding of math concepts. [5][6]. These systems are aligned to formal curricula such as secondary school math and include assessment modules, but they primarily focus on STEM subjects (math in particular) and require specialized materials like tactile sheets or encoded formulas.

Another approach is AI-driven audio-based learning. Audemy (Ying *et al.*, 2025) is an adaptive audio platform designed for blind students. It uses a screen-reader interface and game-like quizzes, adjusting question difficulty, presentation style, and playback speed based on student responses[7]. This personalization assists a customized learning experience and offers instant voice feedback. However, it currently cannot interpret students' emotional cues or nuanced speech inputs[7], and its content is primarily in English, centered on math and games.

Some tools are more limited in scope. For example, REMathEx (Gaura 2002) is a Braille and speech-based tool for reading and editing mathematical expressions [8]. It lets a blind student navigate algebraic formulas via Braille display or speech synthesis, but it is not an interactive tutor – it does not ask or evaluate questions and covers only math notation. Similarly, Agrawal *et al.* (2020) built a voice-controlled exam system that reads multiple-choice questions aloud and records spoken answers[9]. Although useful for taking tests, it only supports fixed-answer questions and does not offer open-ended tutoring. In the same vein, a conceptual model by Almurayziq *et al.* (2022) used Siri and Alexa to let students submit administrative requests via voice[9]; it showed

promise but suffered from inconsistency in matching requests to answers often misinterpreting student queries[9].

There are also AI-powered educational apps for young VI learners. “Animal Watch” (Beal & Rosenblum 2015) is an iPad app that combines AI with Braille and graphics: it presents wildlife-themed math word problems and games to visually impaired children[9]. Students solve nature-related arithmetic problems with spoken guidance, which boosted motivation and skill in tested users. But Animal Watch is narrowly themed focus on animals and the environment, targets lower grade levels, isn’t aligned with broader curricula for higher grades, and lacks a conversational question-and-answer interface.

Platform	Focus on Sri Lankan O/L history syllabus	Student led questions and system led questions	Answer evaluation and correction
audemy	**NO**	**NO**	**YES**
REMathEx	**NO**	**NO**	**YES**
Animal Watch	**NO**	**NO**	**YES**
EduVisionAI	**YES**	**YES**	**YES**

Table 2: Competitive Analysis

1.3 Research Problem

Subject specificity: Almost all reported systems focus on math or basic skills. None address humanities or social-science subjects. For example, many platforms teach math formulas or simple arithmetic [5], and even adaptive tools like Audemy tend to be broad in scope, typically offering drills in math or language arts. To date, no tutor specifically addresses subjects like History or other O-level courses for students with visual impairments.

Answer evaluation: Current tools typically handle constrained answers. Math tutors, for instance, verify numerical steps or multiple-choice answers [5], but they don't assess open-ended explanations. There is little evidence of NLP-based answer checking for free-form questions, which is crucial for subjects like History.

Adaptiveness: Only a few systems personalize their approach to individual learners, such as Audemy which adjusts difficulty levels [6]. Many are fixed-sequence quizzes or manual exercises. Adaptive feedback (beyond right/wrong) and personalization (e.g. tracking long-term progress) are generally weak.

Language and curriculum alignment: most tools are designed in English or adapted to the developers' local context, drawing content from standard English curricula or generic themes, such as Animal Watch, which is focused on ecology [9]. None are designed for Sri Lankan syllabus. For example, an English-only History tutor for Sri Lanka would require entirely new content personalized to the local curriculum.

Interface and cost constraints: Several solutions depend heavily on specialized hardware. Audio-tactile platforms require prepared tactile graphics or expensive "talking tactile tablets". Braille displays are also costly. By contrast, our aim is to develop a voice-only system compatible with common devices, thereby reducing barriers to access.

In summary, while advanced tutoring systems exist for visually impaired students, they are largely math-centric, non-dialogic, or hardware-dependent. Considerably, none offer a voice-based question-answer tutor specifically designed for History or other humanities subjects. A notable gap is the absence of a curriculum-aligned, speech-driven QA tool for high school social studies. This presents a clear research opportunity: to develop an English-language, speech-enabled QA tutor personalized to the Sri Lankan O-Level History syllabus, addressing issues like limited subject coverage, answer evaluation capabilities, adaptability, and accessibility. Such a system would meet an unmet need by providing blind students with an comprehensive, conversational platform to learn history through natural dialogue.

2. OBJECTIVES

2.1 Main Objective

To design and develop an English language, voice-interactive Question-Answering (QA) tutoring system based on AI, trained using the Sri Lankan O-Level History curriculum, to enable visually impaired students to utilize system-initiated mode of questioning with real-time evaluation of answers.

2.2 Specific Objectives

1. **Curriculum Alignment** – Collect, preprocess, and structure Sri Lankan O-Level History syllabus content to train and refine an AI model to provide accurate, context-sensitive responses.

2. **Interactive Voice Interface** – Employ a speech synthesis and recognition module to enable two-way audio interaction.

3. **Answer Evaluation & Feedback** – Integrate natural language processing techniques to evaluate student responses correctness and completeness, producing constructive, syllabus-focused feedback.

4. **Personalized Learning Experience** – Develop adaptive question item selection and difficulty adjustment mechanisms based on learner performance history.

5. **Accessibility Optimization** – Make the platform visually impaired user standards compliant with simple voice commands and minimal visual navigation usage.

6. **Pilot Testing & Validation** – Conduct usability testing and accuracy testing with visually impaired O-Level students in Sri Lanka to evaluate educational impact, system accuracy, and user satisfaction.

3. METHODOLOGY

3.1 Requirement Gathering

The process of requirement gathering involves engaging directly with visually impaired students studying O-Level History to gain a clear understanding of the specific challenges they face when interacting with question-and-answer learning activities.

This includes conducting interviews and focus groups to explore issues related to accessing historical content, formulating questions, providing verbal responses, and receiving audio feedback. We'll also discuss their preferences, whether they favor system-led quizzes or prefer to initiate questions themselves. What's more, we will consult with history teachers to make sure that the system's question bank and evaluation standards are closely aligned with the official Sri Lankan O-Level History curriculum. These discussions will help identify key historical facts, chronology of events, and interpretive skills that should be prioritized in both tutoring and assessment functionalities.

The insights gained from this requirement gathering will guide the development of the AI model, shape the training dataset, and inform the configuration of the speech interface to maximize accessibility, encourage engagement, and ensure educational relevance for visually impaired learners.

3.1.1Past Research Analysis

A review of existing high-tech tutoring systems for visually impaired (VI) learners reveals significant advancements in AI-assisted education but also highlights key limitations in the context of curriculum-specific, voice-interactive learning.

Most prior work has concentrated on STEM-focused assistive tools. For example, audio-tactile mathematics tutors and adaptive learning platforms such as *Audemy* have demonstrated success in delivering subject content through speech interfaces and providing personalized difficulty adjustments. These systems, however, are primarily

designed for numerical or symbolic subjects, with minimal application to humanities or social sciences.

Voice-based examination systems have been developed to facilitate accessible testing for VI students, often using speech synthesis to present multiple-choice questions and speech recognition to record answers. While these approaches improve independent assessment capabilities, they lack open-ended answer evaluation, deeper dialogue capabilities, and domain-specific content understanding.

International research on dialogue-based learning assistants for VI students has shown promise in increasing engagement and comprehension through conversational AI. However, these systems are largely generic knowledge retrieval tools and are not trained on specific curricula, meaning their responses may be incomplete or misaligned with official syllabus requirements.

In the Sri Lankan context, no documented system exists that combines English-only voice-based interaction, curriculum-aligned question generation, and real-time answer validation for the O-Level History syllabus. This gap underscores the need for a tailored solution that merges accessible voice technology with a trained AI model designed to meet both pedagogical accuracy and accessibility requirements for visually impaired learners.

3.2 Development Methodology

The project adopts an Agile development methodology to ensure flexibility, user-centered design, and continuous enhancement of the system's accessibility features. Development is organized into short, iterative sprints, enabling the team to incorporate feedback from visually impaired students and history teachers at every stage.

Initial sprints focus on building the core voice interaction module, enabling speech-to-text input and text-to-speech output for history-related questions. Prototypes are tested with small groups of VI learners to evaluate voice clarity, recognition accuracy, and

response relevance. Feedback gathered during these sessions guides adjustments to voice settings, interaction flows, and question-answer accuracy.

Subsequent iterations expand the system's capabilities by integrating curriculum-specific training data from the O-Level History syllabus and refining the answer validation mechanism to ensure responses are both factually correct and pedagogically aligned. Educator input is continuously sought to maintain syllabus compliance and contextual accuracy.

This iterative cycle of design, testing, and refinement ensures that the system evolves to meet user needs, offering an accessible, reliable, and syllabus-aligned learning experience for visually impaired students.

3.3 Project Management

Project planning and monitoring are managed using Jira, enabling the team to efficiently organize tasks, assign responsibilities, and track progress across all development stages. The project follows a sprint-based workflow aligned with the Agile methodology, ensuring that deliverables are completed within defined timeframes.

Regular sprint reviews are conducted to demonstrate progress, validate completed features, and gather feedback from visually impaired users and educators. Sprint retrospectives are used to reflect on team performance, identify improvement areas, and refine processes for subsequent sprints.

Additionally, Kanban boards and progress tracking dashboards within Jira provide real-time visibility of task statuses, helping the team stay aligned with project objectives, manage dependencies, and address potential delays proactively. This structured yet adaptable approach ensures the project remains on schedule while delivering high-quality, user-focused outcomes.

3.4 Feasibility Study

3.4.1 Technical Feasibility

The proposed system leverages proven and widely available technologies to ensure successful development and deployment.

3.4.1.1 Knowledge on Technologies

To develop the proposed Q&A system for visually impaired learners, knowledge in the following technologies is essential:

- **Natural Language Processing (NLP):** Understanding techniques to train and fine-tune language models for question answering and historical context interpretation.
- **Speech Technologies:** Knowledge of Text-to-Speech (TTS) for delivering answers audibly and Automatic Speech Recognition (ASR) for understanding spoken queries.
- **Machine Learning (ML):** Skills in training, fine-tuning, and evaluating deep learning models, especially transformer-based architectures (LLMs).
- **Data Preprocessing:** Expertise in preparing Sri Lankan O/L history curriculum data, including cleaning, structuring, and converting it into formats suitable for training and testing.
- **Dialogue Systems:** Familiarity with conversational agents and multi-turn dialogue flow for engaging voice-based interaction.
- **Accessibility Design:** Knowledge of designing voice-first interfaces to ensure usability for visually impaired students.
- **Cloud and Deployment Knowledge:** Skills in deploying ML models and APIs using scalable cloud platforms for real-time performance.

Since all these technologies are mature, well-documented, and widely supported, the technical feasibility of the project is high. The team possesses the necessary skills in web/mobile development, accessibility design, and AI integration to implement the system successfully.

3.4.2 Schedule Feasibility

The project is planned over a 6–8 month development cycle, divided into clearly defined phases:

1. **Requirement Gathering & Design** – Collect system requirements from educators, visually impaired students, and accessibility experts; prepare detailed design documents.
2. **Data Collection & Preprocessing** – Gather curriculum-aligned materials, perform OCR optimization, and prepare datasets for AI processing.
3. **Core Module Development** – Implement summarization, text-to-speech (TTS), and speech recognition functionalities.
4. **System Integration** – Combine core modules with the user interface, ensuring accessibility compliance.
5. **Testing & User Feedback** – Conduct iterative usability and performance testing with small groups of target users, refining based on feedback.
6. **Deployment & Maintenance Plan** – Deploy the platform and establish a maintenance schedule.

3.4.3 Economic Feasibility

The project's initial development phase will leverage open-source frameworks and cloud-based services with free-tier or low-cost plans to reduce upfront expenditure. This

approach ensures that core functionalities—such as OCR, summarization, and text-to-speech—can be developed and tested without significant financial burden.

As the platform scales or requires more computational resources for higher accuracy and faster processing, the team will evaluate cost-effective transitions to commercial cloud APIs or dedicated virtual machines. This phased investment strategy allows the project to remain financially sustainable while accommodating growth, enhanced features, and increased user demand.

3.5 System Analysis

3.5.1 Software solution approach

The proposed system adopts a modular software architecture to facilitate scalability, maintainability, and accessibility for visually impaired learners. The core solution integrates three main components:

1. **Voice Interaction Module** – Handles speech recognition for student inputs and text-to-speech (TTS) for system outputs.
2. **Curriculum-Aligned QA Engine** – Implements a trained AI model based on the Sri Lankan O-Level History syllabus. The engine generates relevant answers, validates student responses, and provides real-time feedback, ensuring adherence to syllabus content and pedagogical standards.

The software is designed to operate on standard desktop and mobile devices, using widely supported development frameworks and APIs for speech processing and NLP. A modular, service-oriented design allows each component to be updated or scaled independently, ensuring future adaptability.

By combining voice interaction, AI-based content understanding, and adaptive assessment, this software solution provides an inclusive, interactive, and curriculum-specific learning experience for visually impaired O-Level History students.

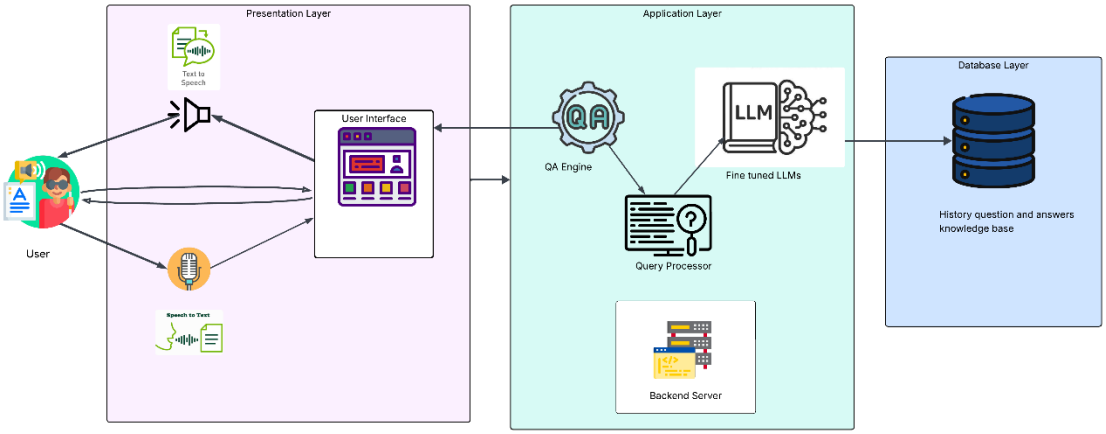

Figure 1: System Diagram

3.5.2 Application of technologies in the relevant area

Research Area: Machine Learning (NLP & Information Retrieval)

The proposed EduVisionAI falls under Machine Learning, specifically Natural Language Processing (NLP) and Question Answering systems, since it requires training models to understand user queries and retrieve accurate answers from history-related content.

Reason for Choosing Machine Learning

1. Understanding Natural Language

Students will ask questions in simple natural language (typed or spoken).

ML-based NLP models can process, interpret, and match these questions to the most relevant answers.

2. Improved Accessibility for Visually Impaired Users

Instead of searching through textbooks, visually impaired students can ask questions directly.

ML enables speech-to-text (STT) and text-to-speech (TTS), improving accessibility.

3. Contextual Accuracy

Traditional keyword-based systems are not effective for complex questions.

ML-based QA models (like LLMs, Transformers, or fine-tuned BERT) can understand the context and provide more meaningful answers.

4. Scalability

Once trained, the system can handle a large database of historical content.

New data (e.g., additional history topics, other subjects in future) can be easily integrated.

5. Personalization

ML can adapt answers to different student levels (slow learners, fast learners) by analyzing previous interactions.

3.5.3 Tools and technologies

Programming Languages:

- **Python:** For NLP model training, preprocessing, and backend logic.
- **JavaScript:** For front-end integration and voice-enabled web interfaces.

Frameworks and Libraries:

- **Hugging Face Transformers:** For fine-tuning pre-trained models on history Q&A tasks.
- **TensorFlow / PyTorch:** For deep learning model development and optimization.
- **NLTK / SpaCy:** For text preprocessing, tokenization, and parsing historical text content.
- **SpeechRecognition (Python) / Vosk:** For ASR (Automatic Speech Recognition).
- **gTTS / Coqui TTS / Microsoft Azure TTS:** For natural and customizable voice-based responses.

Databases:

- **MongoDB:** To store user interaction history and Q&A content in a flexible format.

- **FAISS:** For semantic similarity search, enabling retrieval of relevant answers from the syllabus.

UI and Accessibility Tools:

- **React JS:** For building the student-facing interface.
- **Tailwind CSS:** For responsive and clean UI design.
- **Figma:** For prototyping accessible voice-first UI designs.

Project Development & Management Tools:

- **VS Code / Google Colab:** For code development, experimentation, and testing.
- **Git & GitHub:** For version control and collaborative development.
- **Jira:** For task tracking, sprint planning, and agile project management.

Testing Tools:

- **pytest:** For Python backend testing.
- **Jest:** For front-end React component testing.
- **Postman:** For API testing and validation.

Deployment Tools:

- **Docker:** For containerization to ensure consistency across environments.
- **Kubernetes:** For orchestrating and scaling the deployed system.

3.6 Project requirements

3.6.1 Functional requirements

Voice-Based Interaction – The system must accept student queries via speech and provide spoken responses using text-to-speech synthesis.

Curriculum-Aligned QA – The system should answer questions and validate student responses based on the Sri Lankan O-Level History syllabus.

Quiz Mode – The system must be able to generate questions for the student, accept verbal answers, and evaluate correctness in real-time.

Adaptive Difficulty – The system should adjust question difficulty based on student performance and learning history.

Feedback Generation – The system should provide constructive feedback highlighting errors, key points missed, or clarification suggestions.

Accessibility Features – All functionalities must be usable without visual navigation, supporting screen readers and voice commands.

3.6.2 Non- Functional requirements

Usability – The interface must be simple and intuitive, enabling visually impaired students to interact independently.

Performance – The system should provide real-time responses to student queries with minimal latency.

Scalability – The architecture must allow future expansion, such as adding new subjects or modules.

Reliability – Ensure high availability and accurate speech recognition under diverse accents and voice qualities.

Security – Student data must be securely stored, ensuring privacy and compliance with educational data regulations.

Compatibility – The system should function across desktop and mobile devices, supporting commonly used operating systems.

Maintainability – Code and system modules should be modular to allow easy updates and future enhancements.

3.7 Project scope

3.7.1 In-scope

Development of a voice-interactive Question-Answering system specifically for English-language O-Level History curriculum content.

Implementation of speech recognition for student inputs and text-to-speech (TTS) for system responses.

System-initiated quiz mode where the system asks questions and evaluates spoken answers in real time.

Accessibility optimization to ensure usability for visually impaired students, including voice commands, screen-reader support, and minimal reliance on visual cues.

3.7.2 Out-scope

Development for non-English languages (Sinhala, Tamil) in the QA system; these are handled by other components of the project.

Integration with physical Braille devices or tactile interfaces; the system focuses on voice-only interaction.

Subject areas outside O-Level History, such as mathematics, science, or other humanities subjects.

Manual grading or teacher input for quiz evaluation; the system performs automated assessment only.

Full deployment in multiple schools nationwide during the research phase; initial implementation is limited to pilot testing with a small group of students.

3.8 Evaluation Plan

The testing phase ensures that the **EduVisionAI** meets functional, accessibility, and performance requirements for visually impaired O-Level History students. Testing will be conducted in multiple stages:

3.8.1 Unit Testing

- Individual modules, including speech recognition, text-to-speech (TTS), AI answer evaluation, and adaptive feedback, will be tested separately.
- Unit tests will verify correct functionality, input-output mapping, and handling of edge cases, such as unclear pronunciation or incomplete responses.

3.8.2 Integration Testing

- Test interactions between modules to ensure smooth end-to-end operation, including student queries, system responses, and feedback loops.
- Validate that adaptive difficulty, question sequencing, and progress tracking operate correctly when modules communicate.

3.8.3 Usability Testing

- Conduct with a small group of visually impaired students to assess ease of use, voice command recognition, response clarity, and overall interaction experience.
- Gather qualitative feedback on question pacing, clarity of feedback, and audio interface preferences.

3.8.4 Functional Testing

- Ensure that all functional requirements are met, including quiz mode operation, AI-based answer evaluation, and syllabus alignment.
- Verify that incorrect answers trigger appropriate feedback and hints without causing system errors.

3.8.5 Accessibility Testing

- Confirm compliance with accessibility standards, ensuring full operation without reliance on visual cues and compatibility with screen readers or assistive devices.

3.8.6 Performance Testing

- Measure response times for spoken queries and system-generated feedback to ensure real-time interaction.
- Test system behavior under different accents, speech speeds, and background noise conditions.

The testing phase will conclude with user acceptance testing (UAT), confirming that the system is ready for pilot deployment and meets the educational and accessibility needs of visually impaired learners.

3.9 Timeline

The project will progress through clearly defined phases to ensure systematic development and testing. The initial phase will focus on requirement gathering, including consultations with visually impaired students and history educators to refine system needs. This will be followed by dataset preparation and model fine-tuning to align the QA system with the Sri Lankan O-Level History syllabus. Next, the voice interaction module will be developed, integrating speech recognition and text-to-speech components. System integration will then combine all modules into a functional prototype. The final stages will involve rigorous usability testing with target users, performance evaluation, and refinements based on feedback. The project will conclude with deployment and documentation. A detailed Gantt chart will illustrate the schedule for each phase.

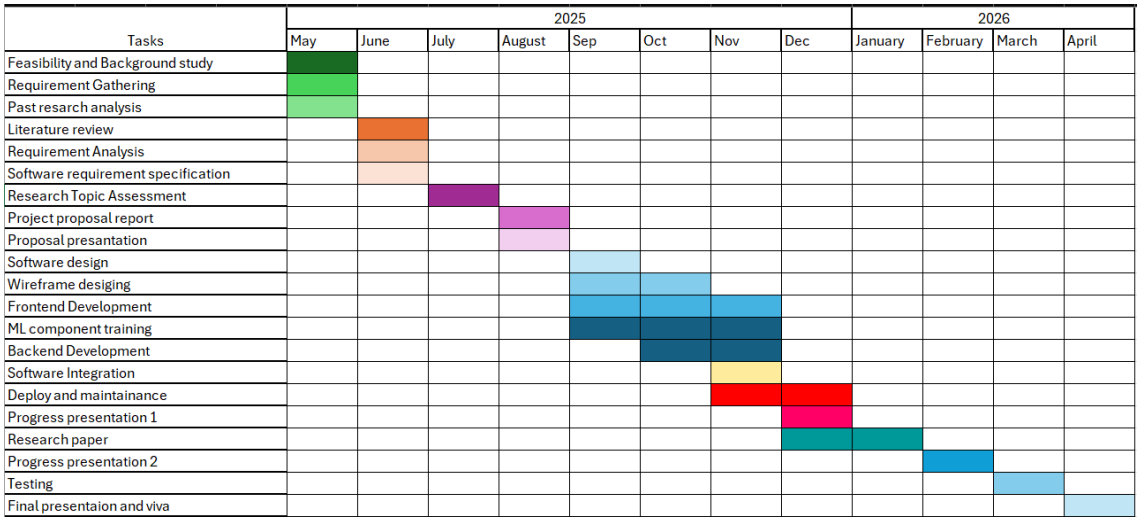

Figure 2: Gantt Chart

3.10 Risk management

Identified Risk	Risk level	Probability of occurrence of the risk	Mitigation strategy
Technical Failures or Bugs in Speech Recognition / TTS	High	High	Seek expert guidance, perform rigorous testing at each development phase, and maintain a bug-tracking system for quick fixes.
Limited Knowledge in AI Model Fine-Tuning for History Content	High	Medium	Engage in targeted online courses, consult with AI specialists, and review academic literature on QA systems for education.

Insufficient Voice Interaction Optimization Skills	High	Medium	Study best practices in voice UI/UX, experiment with existing APIs, and collaborate with accessibility experts.
Difficulty in Gathering Accurate Curriculum-Specific Training Data	High	Medium	Partner with history educators, source materials from official syllabi, and validate all training data for accuracy.
Changes in Project Scope due to Curriculum Updates	High	Medium	Monitor syllabus updates, keep dataset modular for quick adjustments, and consult with supervisors before implementing changes.
Audio Quality Issues Affecting Accessibility	High	Medium	Use high-quality TTS engines, allow adjustable playback speeds, and gather feedback from visually impaired users for improvements.
Data Loss from System or Cloud Failures	High	Medium	Schedule regular backups, use version control for datasets, and employ redundant cloud storage solutions.
User Privacy Risks in Voice Data Handling	High	Low	Anonymize all user voice inputs, use encrypted

			storage, and comply with data protection regulations.
Modifications Requested by Review Panel or Test Users	High	Medium	Address changes promptly, adjust the project plan accordingly, and document all modifications.

Table 3: Risk Management Plan

3.11 Communication management

The goal of the communications management plan is to ensure that the supervisor, co-supervisor, and project team members have timely access to the information needed to perform their roles effectively. Strategic planning and clear communication are essential for the successful completion of the project.

- **Email:** Used for formal updates, sharing documentation, progress reports, and detailed communications that require record-keeping.
- **WhatsApp:** Used for informal, quick exchanges, real-time updates, urgent matters, and brief discussions between team members.
- **Meetings:** Scheduled as needed for in-depth discussions, problem-solving, and decision-making regarding project development.

This plan ensures that all stakeholders remain informed, aligned, and able to contribute effectively, while facilitating smooth coordination between team members and supervisors throughout the project lifecycle.

4. WORK BREAKDOWN

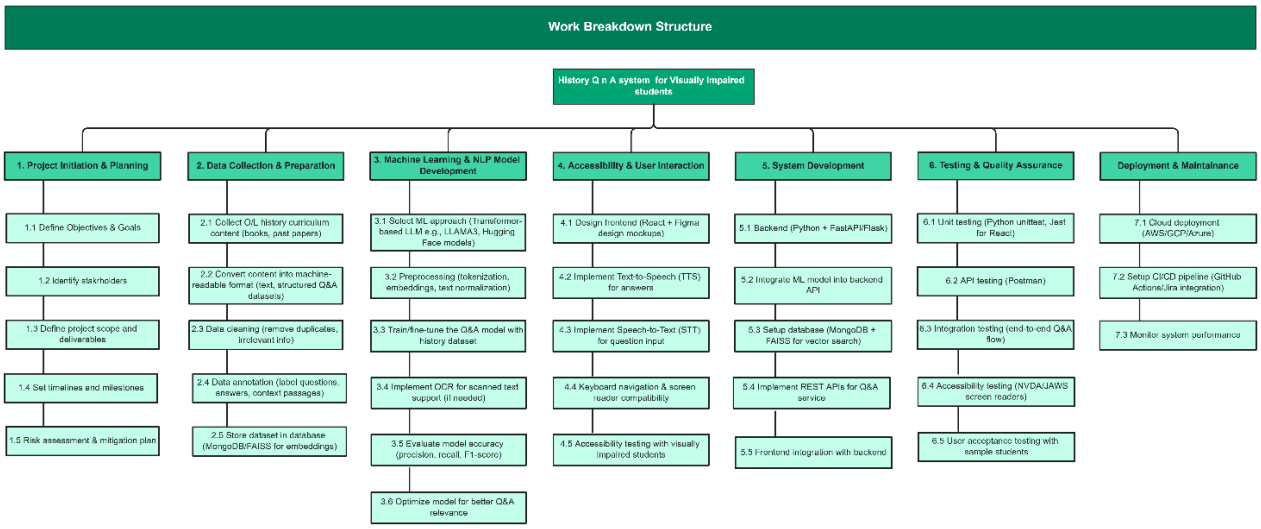

Figure 3: Work breakdown structure

5. COMMERCIALIZATION

The commercialization of the EduVisionAI will focus on generating sustainable revenue while ensuring continued accessibility for visually impaired learners:

1. **Subscription-Based Access** – Schools, institutions, or individual students can subscribe to access the system. Different tiers may be offered, such as:
    - **Basic Tier:** Free access to core features with limited question sets and quiz sessions.
    - **Premium Tier:** Full access to the AI-powered QA engine, adaptive learning features, detailed progress tracking, and additional question banks.
2. **Institutional Licensing** – Partnerships with schools or educational boards to provide bulk access licenses, enabling entire classrooms or schools to use the system in a cost-effective manner. This ensures broader adoption while generating predictable revenue.
3. **Grants and Sponsorships** – Seek funding from organizations focused on **inclusive education**, disability support, or STEM/humanities education initiatives. Sponsorship can subsidize free access for visually impaired learners who cannot afford subscriptions.
4. **Advertising (Optional and Non-Intrusive)** – If appropriate, educationally relevant and non-intrusive advertisements could be incorporated, such as promotions for accessible learning tools, online history courses, or educational resources, ensuring the learning experience remains undisturbed.

This strategy ensures a balanced approach, combining sustainability with accessibility, and enables continuous enhancement of the system's AI capabilities, content updates, and support for visually impaired students across Sri Lanka.

6. BUDGET

To ensure the successful development and implementation of the EduVisionAI, a detailed budget has been prepared to cover all essential project components. The budget accounts for cloud service costs to support AI model training, storage, and hosting, ensuring reliable performance and accessibility. Internet and hosting expenses are included to maintain uninterrupted system availability. Hardware and software tools costs are allocated for development and testing, particularly for speech recognition and text-to-speech integration. Training and workshop expenses are budgeted to familiarize educators and visually impaired students with the system's usage. Publication and dissemination costs will cover promoting the platform through academic publications, awareness programs, and outreach activities. A minimal allocation is also set aside for miscellaneous needs, such as stationery and printing materials for project documentation. The total estimated budget for these components is presented in the table below.

Type	Cost
Internet use and web hosting	10,000
Publication cost	20,000
Testing and training cost	10,000
Other	5000
Total	45,000

Table 4: Cost management plan

7. Summary

This research project aims to develop a voice-based Question-Answering (QA) system tailored to the Sri Lankan O-Level History curriculum, specifically designed for visually impaired students. By integrating speech recognition, text-to-speech synthesis, and AI-powered answer evaluation, the system provides an interactive and accessible learning environment that allows students to engage system-driven questioning.

The project adopts an Agile development methodology, ensuring iterative refinement based on feedback from visually impaired learners and educators. Comprehensive requirement gathering, curriculum-aligned data collection, and AI model training are combined with accessibility-focused design to create a system that is adaptive, accurate, and user-friendly.

The project's scope includes voice-based interaction, AI answer validation, adaptive feedback, progress tracking, and accessibility optimization, while excluding non-English languages, subjects outside O-Level History, and integration with tactile devices. A detailed feasibility study confirms that the project is technically, operationally, and economically viable using open-source tools, cloud platforms, and existing AI frameworks.

Testing will be conducted at multiple levels—unit, integration, usability, and performance—to ensure reliability, real-time response, and accessibility compliance. The commercialization strategy focuses on subscription models, institutional licensing, and potential sponsorship, balancing sustainability with accessibility for target users.

In summary, the proposed system addresses a significant research and educational gap: no existing platform offers a curriculum-specific, voice-interactive, AI-powered History tutor for visually impaired students in Sri Lanka. This project promises to enhance inclusive education, improve learning outcomes, and empower students with autonomous, engaging, and adaptive learning opportunities.

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